The University of Zambia
Department of Mathematics and Statistics
Mat 1110 - Foundation Mathematics and Statistics for Social Sciences

Tutorial Sheet 8- Exponential and Logarithmic Functions **August, 2023.**

1. Sketch the graph of each of the following:

   (a) $f(x) = 5^x$
   (b) $f(x) = -7^x$
   (c) $f(x) = -9^{-x}$
   (d) $f(x) = 2^x + 2$
   (e) $f(x) = 3^x - 4$
   (f) $f(x) = \left(\frac{1}{11}\right)^x$
   (g) $f(x) = -\left(\frac{1}{4}\right)^x$
   (h) $f(x) = 5^{x-1}$
   (i) $f(x) = -11^{x+1}$

2. Solve each of the following equations:

   (a) $2^x = 64$
   (b) $x^3 = 343$
   (c) $\left(\frac{1}{2}\right)^x = \frac{1}{16}$
   (d) $2^x = 0.125$
   (e) $27^{4x} = 9^{x+1}$
   (f) $\left(\frac{1}{8}\right)^{-2t} = 2^{t+3}$

3. Solve each of the following equations:

   (a) $3^{2x} + 3(3^x) - 4 = 0$
   (b) $4^x - 2^{x+1} = 48$
   (c) $4^x - 6(2^x) - 16 = 0$
   (d) $2^{2x+1} - 2^{x+1} + 1 = 2^x$

4. Write each of the following in logarithm form:

   (a) $y = 5^{x+2}$
   (b) $\left(\frac{3}{4}\right)^{-2} = \frac{16}{9}$
   (c) $y = e^x$
   (d) $8^{-1} = 0.125$

5. Write each of the following in exponential form:

   (a) $\log_2 64 = 6$
   (b) $\log_2\left(\frac{1}{32}\right) = -5$
   (c) $\log_{27} 3 = \frac{1}{3}$
   (d) $lne^3 = 3$

6. Evaluate each of the following:

   (a) $\log_3 243$
   (b) $\log_7 343$
   (c) $\log_6\left(\frac{1}{216}\right)$

7. Solve each of the following equations:

(a) $\log_3 x = 4$
(b) $\log_4 x = \frac{3}{2}$
(c) $\log_a 2 = \frac{1}{2}$
(d) $\log_2(3x+4) = 4$
(e) $\ln x = -2$
(f) $e^x = 6$

8. Solve each of the following equations:

   (a) $\log_3 y = 2\log_3 7$
   (b) $\log_5 x - \log_5 4 = 2$
   (c) $\log_x 8 + \log_x 4 = 5$
   (d) $\log_3(x^2+2) = 1 + \log_3(x+2)$
   (e) $\log_4(x^2+8x-1) = 2 + \log_4(x-1)$

9. Sketch each of the following in the indicated interval:

   (a) $f(x) = \log_7 x$
   (b) $f(x) = -\log_{\frac{1}{7}} x$
   (c) $f(x) = -\log_3 x$
   (d) $f(x) = \log_{\frac{1}{3}} x$
   (e) $f(x) = \log(-x)$
   (f) $f(x) = -ln(-x)$
   (g) $f(x) = \ln x + 2$
   (h) $f(x) = 3 - \log_2 x$

10. Sketch the graph of f and g on the same coordinate plane.

    (a) $f(x) = 3^x, \quad g(x) = \log_3 x$
    (b) $f(x) = \left(\frac{1}{2}\right)^x, \quad g(x) = \log_{\frac{1}{2}} x$
    (c) $f(x) = e^x, \quad g(x) = \ln x$

11. A video posted on YouTube initially had 80 views as soon as it was posted. The total number of views to date has been increasing exponentially according to the exponential growth function $y = 80e^{0.2t}$, where t represents time measured in days since the video was posted.

    (a) How many days does it take until 2500 people have viewed this video?
    (b) How many people would have viewed this post in 2 years?

12. Given that the population of certain microorganism is modeled by $P(t) = 1500e^{0.5t}$, where t is time in years. When will the population of these micro-organisms be 3000?

13. Suppose that a stock's price is rising at the rate of 7% per year, and that it continues to increase exponentially. If the value of one share of this stock is ZMK43 now,

    (a) Find the value of one share of this stock three years from now.
    (b) When will one share of this stock be ZMK2000?

14. Money in a certain account is increasing exponentially and is modeled by the formula $C(t) = 100,000e^{kt}$, where t is time in hours. Given that after 1 hour there was K400,000 in this account, how much money will be present after 240 minutes?